

Revised Generation Interconnection System Impact Study Report

For Queue Project Q23-2329

Bellville South 230 kV
February 02, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2329. The Interconnection Customer proposes a 700.0 MW solar photovoltaic facility interconnecting at the Bellville South 230 kV substation in the Illinois area. The request was received on July 03, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 486 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,337,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Weslao – Estill Springs 230 kV line
Contingency: Estill Springs – Elgin Energy Ctr Llc 500 kV
Loading: 184.4% (Rating: 430 MVA)
DFAX: 5.9%
Pre-project loading: 174.8%
2. Facility: Big Oaks – Laureles 100 kV
Contingency: Dekalb – Jones Chapel 500 kV
Loading: 114.8% (Rating: 396 MVA)
DFAX: 5.2%
Pre-project loading: 105.6%
3. Facility: Ward Gulf – Laureles 100 kV
Contingency: Dekalb – Jones Chapel 500 kV line
Loading: 110.1% (Rating: 439 MVA)
DFAX: 5.2%
Pre-project loading: 101.7%
4. Facility: Lions Rd – Aes Beaver Valley Partners Beaver Vly 500 kV
Contingency: Beaver Valley – Cranberry 500 kV
Loading: 107.2% (Rating: 2,652 MVA)
DFAX: 7.9%
Pre-project loading: 105.1%
5. Facility: Cranberry – Beaver Vly 500 kV
Contingency: Aes Beaver Vly Partners Beaver Vly Substation – Lions Rd 500 kV
Loading: 106.7% (Rating: 2,657 MVA)
DFAX: 8.0%
Pre-project loading: 104.6%

6. Facility: Big Oaks Substation – Cypress Crk 100 kV
 Contingency: Jones Chapel – Dekalb 500 kV Line
 Loading: 101.4% (Rating: 422 MVA)
 DFAX: 5.2%
 Pre-project loading: 92.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Weslao – Estill Springs 230 kV line	Rebuild line with higher-capacity conductor	\$12,140,000	19% (\$2,283,000)
Big Oaks – Laureles 100 kV	Rebuild line with higher-capacity conductor	\$2,782,000	7% (\$187,000)
Ward Gulf – Laureles 100 kV	Rebuild line with higher-capacity conductor	\$7,065,000	7% (\$474,000)
Lions Rd – Aes Beaver Valley Partners Beaver Vly 500 kV	Reconductor line	\$46,836,000	7% (\$3,186,000)
Cranberry – Beaver Vly 500 kV	Reconductor line	\$30,208,000	7% (\$2,056,000)
Cypress Crk – Big Oaks 100 kV	Reconductor line	\$2,246,000	7% (\$151,000)
	Total allocated to project		\$8,337,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.